

BADR ALBDULMAJEED

Electrical & Electronic Engineering Graduate

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PROFILE

Fresh Electrical and Electronic Engineering graduate with Honours from Asia Pacific University of Technology & Innovation (APU). Hands-on experience across power systems, protection, PLCs, sensors, embedded hardware, industrial automation, and computer vision, gained through coursework and an R&D role at Control Easy. Final-year project focused on PCB defect detection using a self-supervised Vision Transformer. Currently based in the UAE with a valid Golden Visa and available immediately for a full-time engineering role.

SKILLS & STRENGTHS

- Power Systems & Protection:** Load flow analysis, overcurrent relay coordination, IDMT relays, AC microgrid protection, MATLAB/Simulink, PowerWorld.
- PLC & Industrial Automation:** Siemens S7-1200, S7-300, S7-200 PLCs, TIA Portal, SCL, 24V sensors, relay logic, pneumatic circuits, Industry 4.0 equipment.
- Embedded Systems:** Raspberry Pi 5, NVIDIA Jetson Orin Nano, Arduino, ESP32, GPIO, sensor interfacing, camera integration, optical and relay isolation.
- Computer Vision & AI:** Python, OpenCV, PyTorch, Vision Transformers, YOLOv8, DINOv2, RF-DETR, ONNX Runtime GPU, Roboflow — classification, detection, segmentation.
- Engineering Software:** SolidWorks, TinkerCAD, LTspice, DIALux, LabVIEW, MATLAB App Designer, ROS2.
- Programming:** Python, C, C++, C#, MATLAB, LabVIEW.
- Working Style:** Detail-oriented and clear communicator; learns new tools quickly by testing them and documenting findings.

EXPERIENCE

R&D Project Engineer | Control Easy, Seri Kembangan, Malaysia

Mar 2025 – Jul 2025

- Developed and tested machine-vision applications for industrial quality control using classification, object detection, and segmentation.
- Automated inspection:** built a golf-ball inspection station using a Raspberry Pi 5, proximity sensor, camera, and Vision Transformer model for real-time classification and sorting.
- Hardware interfacing:** connected a 24V industrial proximity sensor to 3.3V Raspberry Pi GPIO using optical and relay isolation, then tested the complete hardware and software setup.
- PLC integration:** integrated vision systems with PLC-controlled Industry 4.0 equipment; received hands-on training with Siemens S7-1200, S7-300, and S7-200 PLCs using SCL in TIA Portal.
- Technical documentation:** prepared manuals covering dataset creation, model training, GUI deployment, hardware setup, and electrical wiring.

SELECTED PROJECTS

Final Year Project — Self-Supervised Vision Transformer for PCB Defect Detection

PyTorch · SimCLR · MaxViT-Small · RF-DETR · DINOv2 · ONNX Runtime CUDA · Streamlit · OpenCV · Roboflow

Built a PCB inspection system achieving 77.50% test accuracy, 0.8281 weighted F1 across 22 classes, 68.83% mAP@50, and 37.25 FPS after GPU optimisation.

Hospital Logistics AGV Vision System

NVIDIA Jetson Orin Nano · ZED 2i · YOLOv8 · PyTorch · ONNX Runtime GPU · ROS2 · PyQt5 · Roboflow

Developed the vision subsystem for an autonomous hospital logistics robot, achieving 98.99% mAP@50, 98.66% precision, and 97.85% recall.

Overcurrent Relay Coordination Tool

MATLAB App Designer

Built a tool for IDMT relay coordination, operating-time calculations, Time-Current Characteristic curves, system analysis, and validation reporting for ten overcurrent relays; maximum calculation error 1.00%.

10-Bus Power Flow Analysis

MATLAB — Newton-Raphson, Gauss-Seidel, Fast Decoupled methods

Built and analysed a 10-bus system, evaluating bus voltage, line flow, generation, and losses under additional loading.

AC Microgrid Overcurrent Protection

MATLAB/Simulink

Modelled a 3-bus AC microgrid with transformers, distribution lines, CTs, circuit breakers, and overcurrent relays; tested IEC and IEEE inverse relay characteristics under different fault types.

OTHER NOTABLE PROJECTS

- **Medication Scanner & Obstacle Detection GUI (Python, Jetson):** Python GUI on Jetson for medication image scanning and obstacle detection.
- **Modular Smart Power Hub — The Honeycomb System (SolidWorks):** Modular power distribution hub with a honeycomb-inspired structure.
- **Automated Golf Ball Quality Control (Raspberry Pi, Vision Transformer):** Camera-based inspection setup classifying golf balls in real time.
- **PCB Component Inspection System (OpenCV, AI):** AI-based image processing to inspect PCB components and identify visual defects.
- **Wood Defect Segmentation System (Computer Vision):** Segmentation workflow to locate and classify visible defects in wood samples.
- **Fabric Quality Control System (Computer Vision):** Workflow to inspect fabric images and identify quality issues.
- **Vaccination Center System (Python):** System to organise vaccination-centre records and basic service flow.
- **Vaccination Inventory System (C++):** Inventory system to track vaccine stock, records, and updates.
- **Coffee Making Process (LabVIEW):** Process model for monitoring and controlling coffee preparation steps.
- **Automated Bus Door Control (PLC & Pneumatics):** PLC and pneumatic control sequence for safe automated bus-door operation.
- **Automotive Part Control System (C):** Control system for handling and checking automotive parts.
- **Library Lighting Design (DIALux):** Planned library lighting levels and fixture layout.
- **Automated Parking System (Arduino):** Vehicle detection and space control for a parking system.
- **Sound Purifier (MATLAB):** Signal-processing model to reduce unwanted sound.
- **BJT Audio Amplifier (MATLAB):** Simulated and analysed a BJT amplifier for audio signal amplification.
- **Discrete-Time System using Z-Transform:** Analysed a discrete-time system using Z-transform methods and MATLAB.
- **Temperature Control System (Arduino, TinkerCAD):** Model for temperature sensing and automatic control.
- **AI Hand Gesture Recognition (Python, OpenCV):** System to recognise hand gestures from camera input.
- **Vision-Based Fruit Inspection System (Python, OpenCV, Roboflow):** Inspected fruit images for visual quality classification.
- **Single-Phase Transformer & Three-Phase Alternator Simulation:** Simulated transformer and alternator behaviour; reviewed voltage/current characteristics.
- **FM Modulation and Demodulation (LabVIEW):** Model for FM signal generation, modulation, and demodulation.
- **Microgrid Technology Simulation (MATLAB, LabVIEW):** Simulated microgrid operating concepts.
- **Off-Grid Wind Energy System (Simulink):** Modelled an off-grid wind energy system and reviewed its output.
- **Go-Kart Design (SolidWorks):** Designed a go-kart assembly and mechanical components.

EDUCATION

Bachelor of Electrical and Electronic Engineering with Honours Completed 2026

Asia Pacific University of Technology & Innovation (APU) — 2020–2022; withdrew; rejoined 2024–2026 · Fresh Graduate

American High School Diploma 2017 – 2019

Sama American School — Cairo, Egypt

COURSES & CERTIFICATES

- **MATLAB Onramp** — MathWorks (Sep 2026)
- **Simulink Onramp** — MathWorks (Sep 2026)
- **Core MATLAB Skills Learning Path** (MATLAB Desktop Tools & Troubleshooting Scripts, Explore Data with MATLAB Plots, Make and Manipulate Matrices, Calculations with Vectors and Matrices) — MathWorks (Sep 2026)

LANGUAGES

- **Arabic** — Native
- **English** — Fluent